

Semiclassical transition state theory. A new perspective

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The semiclassical transition state theory (SCTST) introduced by Miller, Hernandez, Handy, Jayatilaka and Willetts requires the inversion of an (effectively integrable) Hamiltonian with respect to the action of the reactive coordinate. It is shown that the inversion may be avoided in computing the thermal rate constant; the resulting expression also provides an appealing link to conventional transition state theory. This reformulation of the SCTST rate is illustrated by application to the bimolecular reaction, $\text{H} + \text{H}_2 \rightarrow \text{H}_2 + \text{H}$, and to the unimolecular dissociation, $\text{D}_2\text{CO} \rightarrow \text{D}_2 + \text{CO}$.

1. Introduction

The thermal rate constant for a chemical reaction can be conveniently expressed as a Boltzmann average of the cumulative reaction probability (CRP) $N(E)$ [1–3],

$$k(T) = (2\pi\hbar Q_r)^{-1} \int_{E_0}^{\infty} dE e^{-\beta E} N(E), \quad (1)$$

where Q_r is the reactant partition function (per unit volume), E_0 is the quantum mechanical reaction threshold and $\beta \equiv (kT)^{-1}$ is proportional to the inverse temperature as usually defined. The microcanonical rate, typically of most interest for unimolecular reactions, is also given in terms of the CRP

$$k(E) = [2\pi\hbar\rho_r(E)]^{-1} N(E), \quad (2)$$

where ρ_r is the reactant density of states (per unit energy).

Considerable effort has been focused recently on developing numerically efficient methods to evaluate the CRP exactly (see, e.g., ref. [4]), but at a much simpler level we have shown how to implement a semiclassical transition state theory (SCTST) approximation for the CRP [5,6]. Because SCTST is much easier to implement than more rigorous numerical approaches, it can be more readily applied

to complex reactions. Though this SCTST clearly contains the basic transition state approximation of “direct dynamics”, it nevertheless includes the effect of non-separable couplings between all degrees of freedom – including the reaction coordinate – in the transition state region. We also note that SCTST is similar to the most general form of variational TST [7] in that it corresponds [8] to computing flux through a dividing surface in the full phase space of the system; the commonly used version [9] of variational TST only considers dividing surfaces in coordinate space.

The purpose of this Letter, is to show that the SCTST expression for the thermal rate constant, $k(T)$, can be cast in a much more useful form, one that is simpler to implement than the standard theory already developed and one that can more readily incorporate dynamical extensions to the theory. This new form of the SCTST thermal rate also has an appealing conceptual link to standard TST.

The next section briefly summarizes SCTST and recasts the result for $k(T)$ into the new form. Various aspects of this new rate expression are illustrated in section 3 by application to several representative reactions.

2. Methodology

2.1. Summary of semiclassical TST

The starting point for semiclassical TST is to note [8] that the classical Hamiltonian can in general be expressed in terms of a set of locally conserved ("good") action variables associated with the transition state (i.e. saddle point) region of the potential energy surface. The first step in SCTST is thus to determine the classical Hamiltonian $\mathcal{H}^{\text{cl}}(I) \equiv \mathcal{H}^{\text{cl}}(I_1, \dots, I_F)$ in terms of the F actions $\{I_k\}$ (where F is the number of degrees of freedom.) The $F-1$ actions associated with the bound degrees of freedom are quantized in the usual semiclassical (Bohr-Sommerfeld) fashion,

$$I_k = (n_k + \frac{1}{2})\hbar, \quad \text{for } k=1, \dots, F-1, \quad (3)$$

where $n^* \equiv \{n_k\}$ are the quantum numbers for states of the activated complex. The action I_F – the one associated with the reaction coordinate – is pure imaginary and defines the generalized barrier penetration integral^{*1} θ ,

$$\theta \equiv -i\pi I_F / \hbar. \quad (4)$$

θ is determined as a function of total energy E and the $F-1$ quantum numbers of the activated complex by energy conservation

$$\mathcal{H}^{\text{cl}}[I_k = (n_k + \frac{1}{2})\hbar, I_F = i\hbar\theta/\pi] = E, \quad (5)$$

i.e. for n^* fixed, one must invert the E - θ relation defined by eq. (5) to obtain $\theta(E, n^*)$. Since the dynamics is integrable in terms of the "good" actions, the transmission probability for state n^* and energy E has the same form as in one dimension, i.e. $(1 + e^{2\theta})^{-1}$, so the CRP is given by [8]

$$N(E) = \sum_{n^*} \{1 + \exp[2\theta(E, n^*)]\}^{-1}. \quad (6)$$

Note that while in this discussion the semiclassical Hamiltonian has been constructed from the classical Hamiltonian, it is also possible to obtain it from the quantum Hamiltonian $\mathcal{H}^{\text{qm}}(n)$ if one is able to determine the latter in terms of a complete set of good quantum numbers n . (This connection has recently been made more explicit by the demonstration that

the semiclassical energies of the activated complex correspond to the (Siebert) eigenvalues of the complex Hamiltonian with imposed outgoing wave boundary conditions [11]. The generalized barrier penetration integral is defined in analogy with eq. (4) as

$$\theta \equiv -i\pi(n_F + \frac{1}{2}), \quad (7)$$

where n_F is the vibrational "quantum number" associated with the reactive mode. Because this mode is not bound, θ is again a continuous variable and may be obtained by inverting the Hamiltonian $\mathcal{H}^{\text{qm}}(n^*, n_F)$,

$$\mathcal{H}^{\text{qm}}[n^*, n_F = (i\theta/\pi - \frac{1}{2})] = E. \quad (8)$$

It may appear that this procedure is difficult to implement because of the difficulty associated with obtaining the quantum Hamiltonian $\mathcal{H}(n^*, \theta)$. In fact, $\mathcal{H}(n^*, \theta)$ may be easily obtained using perturbation theory as will be described in section 2.3.

2.2. The "new" semiclassical rate expression

Eq. (6) is the original expression obtained for the CRP in the SCTST approximation. We now proceed to use it in eq. (1) for the thermal rate,

$$k(T) = (2\pi\hbar Q_r)^{-1} \sum_{n^*} \int_{E_0}^{\infty} dE \exp(-\beta E) \times \{1 + \exp[2\theta(E, n^*)]\}^{-1}, \quad (9)$$

where we have interchanged the order of summation and integration. Since one must integrate over all E in eq. (9), it is convenient to change the integration variable from E to θ and thus avoid having to invert the E - θ relation. (The analysis here assumes that the E - θ relation is one-to-one; the argument is modified somewhat if this is not the case but the form of the result is unchanged [12].) Thus

$$\begin{aligned} & \int_{E_0}^{\infty} dE \exp(-\beta E) \{1 + \exp[2\theta(n^*, E)]\}^{-1} \\ &= \int_{-\infty}^{\infty} d\theta \frac{\partial E(n^*, \theta)}{\partial \theta} \exp[-\beta E(n^*, \theta)] \\ & \times [1 + \exp(2\theta)]^{-1} \end{aligned} \quad (10)$$

*1 The one-dimensional barrier penetration is defined, e.g., in ref. [10].

$$= \frac{1}{\beta} \int_{-\infty}^{\infty} d\theta \exp[-\beta E(n^*, \theta)] \frac{1}{2} \operatorname{sech}^2(\theta), \quad (11)$$

where the last line results from an integration by parts. (The surface terms vanish because $E(\theta \rightarrow \infty) = E_0$, the reaction threshold, and $E(\theta \rightarrow -\infty) = \infty$.) Use of eq. (11) in eq. (9) then gives

$$k(T) = (2\pi\hbar Q_i \beta)^{-1} \int_{-\infty}^{\infty} d\theta \frac{1}{2} \operatorname{sech}^2(\theta) \times \sum_{n^*} \exp[-\beta E(n^*, \theta)], \quad (12)$$

where we have again changed the order of summation and integration. (Note that no issues of convergence with respect to this reordering need arise since it can be avoided in a more detailed derivation [12].) Noting that $(2\pi\hbar\beta)^{-1} = kT/h$, eq. (12) takes the form of the traditional TST,

$$k(T) = \frac{kT}{h} \frac{Q^*(T)}{Q_i(T)}, \quad (13a)$$

by defining the reactive partition function of the activated complex (including the tunneling correction factor which is not separable from it) as an average of the *pre-reactive partition function* $Q^*(T, \theta)$ with the weight function $\frac{1}{2} \operatorname{sech}^2(\theta)$,

$$Q^*(T) = \int_{-\infty}^{\infty} d\theta \frac{1}{2} \operatorname{sech}^2(\theta) Q^*(T, \theta), \quad (13b)$$

where

$$Q^*(T, \theta) = \sum_{n^*} \exp[-\beta E(n^*, \theta)]. \quad (13c)$$

Eq. (13) is the basic theoretical result of the Letter. Since $E(n^*, \theta) = \mathcal{H}(n^*, \theta)$ is simply the semiclassical Hamiltonian, there is now no need to invert the E - θ relation (as is necessary to determine the semiclassical CRP.) For applications to date (vide infra) based on second-order perturbation theory, $\mathcal{H}(n^*, \theta)$ has been a quadratic function of θ so the inversion has been no problem, but if one were to use higher-order perturbation theory – or even non-perturbative treatments that are necessary when Fermi resonance-like interactions exist – it is a major advantage not to have to obtain $\theta(n^*, E)$ ex-

plicitly. Along this line, we note that eq. (13c) is the semiclassical equivalent to

$$Q^*(T, \theta) = \operatorname{Tr} e^{-\beta \mathcal{H}(\theta)}, \quad (14)$$

which shows that it is not even necessary to quantize the $F-1$ degrees of freedom of the activated complex explicitly but only to determine their partition function as a function of the action, θ , in the reaction coordinate. This opens the door to the possibility of even more accurate ways of determining $Q^*(T, \theta)$ that go beyond the treatments employed so far.

2.3. Construction of the Hamiltonian, $\mathcal{H}(\theta)$

For a general multi-dimensional system the classical Hamiltonian will seldom be integrable, and even if it is, it may not be obvious how to obtain the corresponding good global action-angle variables. A standard procedure to avoid these difficulties is to characterize the local dynamics using perturbation theory in the local region of interest^{#2}. Since the rate expression derived in section 2.2 focuses on the transition state region, it is natural to expand the Hamiltonian about the saddle point and use perturbation theory to successively obtain an approximate Hamiltonian to an order consistent with the expansion.

As described in section 2.1, the semiclassical Hamiltonian may also be obtained from the quantum Hamiltonian. Perturbative methods already developed to describe rovibrational eigenvalue problems [14], i.e. stationary minima, can be readily adapted to describe the transition state region, i.e. saddle points. The application of canonical Van Vleck perturbation theory (CVPT)^{#3} to the transition state region is discussed in detail in ref. [6] for the vibrational Hamiltonian and in ref. [16] including angular momentum J . In this section, we briefly review the methodology for $J=0$.

The vibrational Hamiltonian expanded in mass-weighted normal mode coordinates about the saddle point is

^{#2} For a review of the classical perturbation theory using canonical transformations, see ref. [13].

^{#3} For a discussion of the quantum mechanical CVPT, see ref. [15].

$$\mathcal{H} = V_0 + \frac{1}{2} \sum_k^F \left(-\hbar^2 \frac{\partial^2}{\partial Q_k^2} + \omega_k^2 Q_k^2 \right) + \frac{1}{6} \sum_{klm}^F f_{klm} Q_k Q_l Q_m + \frac{1}{24} \sum_{klmn}^F f_{klmn} Q_k Q_l Q_m Q_n, \quad (15)$$

where V_0 is the potential energy at the saddle point, $\{f_{klm}\}$ ($\{f_{klmn}\}$) are the cubic (quartic) derivatives at the saddle point, and the sums over lowercase letters are unrestricted sums over the F vibrational modes. The difference from the standard vibrational expansion [14] is that at zeroth order the potential along the reaction coordinate (mode F) is a harmonic barrier so that the frequency ω_F is imaginary,

$$\omega_F = i|\omega_F| \equiv i\tilde{\omega}_F. \quad (16)$$

Using CVPT at second order, the Hamiltonian is given in terms of the "good" quantum numbers by

$$\mathcal{H} = V_0 + \sum_{k=1}^s \hbar\omega_k(n_k + \frac{1}{2}) + \sum_{k \leq k'}^s \hbar x_{kk'}(n_k + \frac{1}{2})(n_{k'} + \frac{1}{2}), \quad (17)$$

where the ω and x are the "spectroscopic constants" describing the transition state. The resulting E - θ relation may be written explicitly as

$$E(n^*, \theta) = V_{n^*} - \hbar\omega_{n^*}^* \left(\frac{\theta}{\pi} \right) - \hbar x_{FF} \left(\frac{\theta}{\pi} \right)^2, \quad (18)$$

where

$$V_{n^*} = V_0 + \sum_{k=1}^{F-1} \hbar\omega_k(n_k + \frac{1}{2}) + \sum_{k \leq k'}^{F-1} \hbar x_{kk'}(n_k + \frac{1}{2})(n_{k'} + \frac{1}{2}), \quad (19)$$

$$\omega_{n^*}^* = -i\omega_F + i \sum_{k=1}^{F-1} x_{kF}(n_k + \frac{1}{2}). \quad (20)$$

Note that V_{n^*} and $\omega_{n^*}^*$ are real numbers because both the anharmonic coupling $\{x_{kF}\}$ between the perpendicular and reactive modes, and the reactive mode frequency ω_F , are imaginary. The transitions from the f to the x can be found in ref. [6]. While only second-order perturbation theory has been described explicitly, the generalization to higher-order perturbation theory using CVPT [17] (taking care

of the imaginary terms) follows readily.

The thermal rate can now be obtained by substitution of eq. (18) into the expressions developed in section 2.2. However, if there exists a resonance in the expansion at second order, then the corresponding coupling terms cannot be removed perturbatively [14]. These terms must be retained in the zeroth-order Hamiltonian and the remaining coupling terms are successively removed using perturbation theory. The resulting Hamiltonian is block diagonal, characterized by resonance polyads describing each set of resonating states. This Hamiltonian will remain diagonal with respect to θ because the frequency corresponding to the reactive mode is imaginary and consequently cannot cause a small resonance denominator. Thus the Hamiltonian parametrized by θ can be diagonalized and use of eq. (13) provides the SCTST thermal rates.

In analogy with the use of diagonalization of the resonance polyads, one can instead retain a large part of the vibrational Hamiltonian at zeroth order, and still be able to use the rate expressions in section 2.2. The only good action that needs to be identified in the *pre-reactive* partition function is the action, or quantum number n_F , associated with the reaction coordinate. Thus one only needs to use perturbation theory to identify this good action in the Hamiltonian. The remaining off-diagonal couplings can be treated at zeroth order and the pre-reactive partition function calculated using a full Hamiltonian matrix. For large systems in which only a few modes are strongly coupled to the reaction, the weakly coupled modes may still be treated perturbatively thereby reducing the dimensionality of the pre-reactive partition function calculation. The use of these *mixed-diagonalization* methods has yet to be explored.

3. Illustrative examples

3.1. Eckart barrier

As a test of the semiclassical expressions presented above, we first consider the one-dimensional Eckart barrier for which $\mathcal{H}(\theta)$ can be obtained exactly. The symmetric Eckart barrier [18] can be written as

$$V(x) = V_0 \operatorname{sech}^2(\pi x/a), \quad (21)$$

where the parameters V_0 and a characterize the barrier height and length scale, respectively. Evaluation of the barrier penetration integral (see footnote 1)

$$\theta(E) = \int dx \sqrt{2m[V(x) - E]}/\hbar, \quad (22)$$

leads to

$$\theta(E) = \sqrt{\frac{2m}{\hbar^2}} a (\sqrt{V_0} - \sqrt{E}), \quad (23)$$

for $0 \leq E < V_0$. The restriction for $E < V_0$ is removed by appealing to the uniformly convergent semiclassical theory [19]. Inversion of eq. (23) results in a Hamiltonian $E(\theta)$, which is identical in form to that obtained by perturbation theory, eq. (18), except that here the expression is exact. This system thus provides a direct test of the semiclassical approximation involved in the SCTST theory.

The pre-reactive partition function for the Eckart barrier is

$$Q^*(T, \theta) = \exp\{-\beta[V_0 - \hbar\omega^*(\theta/\pi) - \hbar x^*(\theta/\pi)^2]\}, \quad (24)$$

where

$$\omega^* = \left(\frac{2V_0}{ma^2}\right)^{1/2}, \quad (25)$$

$$x^* = -\frac{\hbar}{2ma^2}. \quad (26)$$

The integral over θ in eq. (13b) then gives

$$k(T) = \Gamma^* \frac{kT \exp(-\beta V_0)}{h Q_r}, \quad (27)$$

where the tunneling correction factor Γ^* is

$$\Gamma^* = \sum_{n=0}^{\infty} \left(\sum_{m=0}^n \frac{(\beta \hbar \omega^*)^{2m} (\beta \hbar x^*)^{n-m}}{(2m)!(n-m)!} \right) b_n, \quad (28)$$

where

$$b_n \equiv (-1)^{n+1} (1 - 2^{1-2n}) B_{2n}, \quad (29)$$

and B_{2n} are Bernoulli numbers.

It is not hard to show that eq. (28) reduces to the standard harmonic expression $\frac{1}{2} \hbar \omega^* / \sin(\frac{1}{2} \hbar \omega^* \beta)$ in the limit that the anharmonic constant $x^* \rightarrow 0$. In this limit Γ^* is finite only for $\beta < 2\pi$, while it is finite for all temperatures if $x^* \neq 0$. To show the level of accuracy that can be expected of SCTST, fig. 1 shows

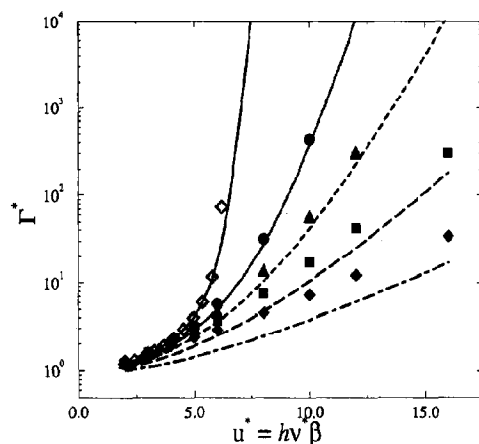


Fig. 1. The tunneling correction factor Γ^* for the transmission through Eckart barriers for various relative anharmonicities $|x^*/\omega^*|$. Exact results [21] (points) are compared to the SCTST result (curves). The dotted curve ($\bullet$) corresponds to $|x^*/\omega^*| = \pi/32$. The short-dashed curve ($\blacktriangle$) corresponds to $|x^*/\omega^*| = \pi/16$. The long-dashed curve ($\blacksquare$) corresponds to $|x^*/\omega^*| = \pi/8$. The dot-dashed curve ($\blacklozenge$) corresponds to $|x^*/\omega^*| = \pi/4$. The solid curve ($\diamond$) corresponds to $x^*/\omega^* = 0$ - i.e. the harmonic limit - which diverges at $\beta = 2\pi$.

the SC result for Γ^* as a function of temperature given by eq. (28), compared to the exact quantum results reported by Johnston [20] for various ratios of $|x^*/\omega^*|$. The accuracy of the SCTST result is extremely good up to anharmonicities $|x^*/\omega^*|$ of $\approx 20\%$ ($\approx \pi/16$), and only about a factor of 2 too small by the time it has increased to $\approx 80\%$ ($\approx \pi/4$). Thus for systems of chemical interest it is reasonable to expect that the rate expressions in section 2 will provide good estimates of the thermal rates in a wide thermal range.

3.2. $H + H_2 \rightarrow H_2 + H$

The transmission probabilities and thermal rates for the distinguishable $H + H_2$ reaction with zero angular momentum have recently been obtained using the SCTST discussed in section 2.1 [6]. For this reaction, $\mathcal{H}(n^*, \theta)$ is constructed using the perturbation theory described in section 2.3. The cubic and quartic derivatives at the saddle point were obtained by finite differences of the DMBE [21] (direct many-body expansion) surface and these can be transformed to provide the "spectroscopic constants"

characterizing the transition state region [6]. Since the out-of-plane degenerate bends are in Fermi resonance with the symmetric stretch, it is necessary to use the degenerate form of perturbation theory. The perturbative Hamiltonian, $\mathcal{H}(\theta)$, is block diagonal with each block labeled by the polyad quantum number of the degenerate modes. In the calculation of the CRP the required inversion of the $E-\theta$ relations must be performed numerically since one must diagonalize each of these blocks as a function of θ . (In fact, in previous work [6] this inversion was approximated using a quadratic form determined by the eigenvalues of the Hamiltonian at $\theta=0$.) However, the θ -dependent anharmonic coupling may be easily included without requiring this inversion by using the SCTST thermal rate expression developed in section 2.

In fig. 2, the semiclassical anharmonic thermal rate for the $\text{H}+\text{H}_2$ reaction is compared both to the harmonic rate and to the exact rate [22] on the DMBE surface. The semiclassical rates are on the order of 70% of the exact values at all temperatures, and they converge well below the harmonic limit of $\hbar\omega^*\beta=2\pi$, $T=323$ K for this system.

The pre-reactive partition functions at various sample temperatures are shown in fig. 3. As can be seen, they peak in the positive θ region and broaden with increasing temperature. The broadening results from the increasing importance of non-tunneling transmission, while the peaking in this system results from the strong anharmonicity. The integrands of the semiclassical reactive partition function in eq. (13b) are also shown and they exhibit even stronger peak-

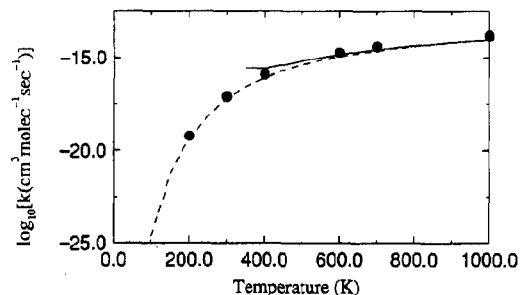


Fig. 2. The thermal rate for the $\text{H}+\text{H}_2$ reaction on the DMBE surface. The standard harmonic result (solid) and the anharmonic semiclassical rate (dashed) are compared to the exact results of Chatfield et al. [22] (points).

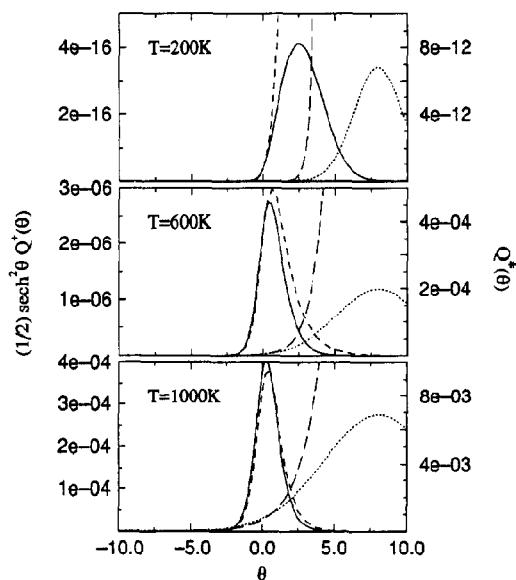


Fig. 3. SCTST results for the $\text{H}+\text{H}_2$ reaction at various temperatures. The long-dashed (dotted) curves are the harmonic (anharmonic) pre-reactive partition functions $Q^*(\theta)$ with the scaling determined by the axes on the right. The short-dashed (solid) curves are the harmonic (anharmonic) integrands of eq. (13b) with the scaling determined by the axes on the left.

ing around $\theta=0$. The peaking manifests itself in the convergence of the rate, while the asymmetry in the integrand results from tunneling effects. As the temperature increases, tunneling is less significant, and the peaks in the integrand are narrower and more symmetric.

3.3. $\text{D}_2\text{CO} \rightarrow \text{D}_2 + \text{CO}$

At temperatures for which there is a non-negligible rate for the dissociation of formaldehyde D_2CO , the dynamics takes place on the excited potential energy surfaces, T_1 and S_1 , in addition to the lowest-lying surface S_0 [23]. Since the theory outlined in section 2 does not account for multiple potential energy surfaces nor for non-adiabatic effects, here we focus on the calculation of the reactive partition function making the non-physical assumption that the dynamics occurs exclusively on the S_0 surface. This should illustrate the feasibility of the method for unimolecular dissociation.

The "spectroscopic constants" for the D_2CO dis-

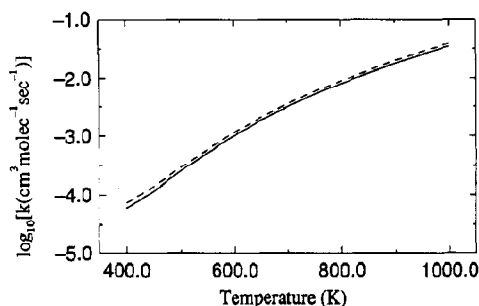


Fig. 4. The reactive partition function for the formaldehyde dissociation for the harmonic case (solid) is compared to the anharmonic case (dashed).

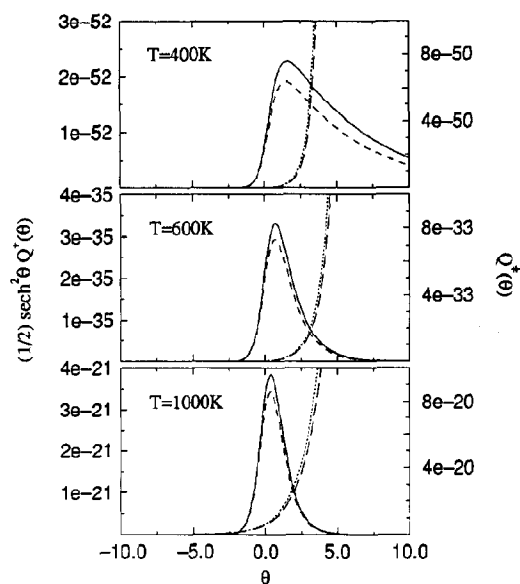


Fig. 5. SCTST results for the D_2CO dissociation. The curves are as in fig. 3.

sociation have been determined using the derivatives of the surface calculated by second-order Møller-Plesset perturbation theory (MP2/DZP) [5]. The reactive partition function, eq. (13b), can be readily obtained and is shown in fig. 4. As has already been noted [24], the anharmonic effect on the reactive partition function is small. Unfortunately, the anharmonicity of the reactive mode is also small and, at temperatures below the harmonic cutoff – 400 K – is not sufficient to ensure the convergence seen in the $H+H_2$ reaction.

In fig. 5, the pre-reactive partition function, and the integrand for the semiclassical rate (see eq. (13)) are shown. At the temperatures presented, the thermal rate converges although it should be clear that the tail of the integrand is rising with decreasing temperatures; at still lower temperatures, this tail is not damped and results in the diverging integral. The use of higher-order perturbation theory could provide stronger damping terms, and result in convergence at these lower temperatures. This conjecture has yet to be explored.

4. Concluding remarks

The primary theoretical development of this Letter has been the determination of a new form for the semiclassical rate expression allowing for the inclusion of anharmonicity (either through perturbation theory or mixed-diagonalization) using a direct analysis of the local Hamiltonian in the transition state region. The pre-reactive partition function, defined in the Letter, is readily computed in analogy to the calculation of standard partition functions and offers a new tool for the analysis of the convergence of the thermal rate. Moreover, anharmonic effects can actually lead to convergence at low temperatures where the harmonic rate diverges.

These conclusions have been illustrated for the $H+H_2$ reaction, where the strong anharmonicity of the transition state converges the rate at low temperatures. The presence of a Fermi resonance between some of the modes did not present a difficulty because the new thermal rate expression focuses directly on the eigenvalues of the θ -labeled Hamiltonian.

In the formaldehyde dissociation, at second order in perturbation theory, the anharmonic semiclassical rate expression gave a small correction to the harmonic result; but it did not extend the region of convergence. This latter failing is attributed to the small anharmonic coupling of the reaction coordinate at this order in perturbation theory. It is possible that this result can be improved by using still higher-order perturbation theory or mixed-diagonalization.

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